

Many happy returns?

THIS IS A MONTH OF RETURNS, to be honest. After Siddharth left, I've returned to a more hands-on role for the magazine – much to the dismay of the team I should add. Sid's sorely missed, but we're glad he's happier on his quest to rediscover himself. Aditya returned to us after many years, and you will see more articles from him from next month onwards. Plus, we've returned to a topic that's been in and out of favour with the technology world for my entire career with Digit – AI. Like I said, a month of many returns!

AI is something that filled me with disgust, or with hope, or with excitement, depending on which year you asked me for my reaction. If anything, the sheer amount of times I've seen it being hyped up in my 20 years in tech has made me a little jaded. I seem to take even the most wonderous sounding advancements with a fistful of salt (because a pinch has proven too little in the past).

So am I trying to suggest that AI isn't going to happen? Not at all, I honestly think it's a question of when, and not if. It's just that I'm not as excited when small advances are presented as breakthroughs. In fairness, the job of most marketing people is to do just that, and that's why AI will be the buzzword of 2018, without a doubt. However, I think it's all just a polishing up and presenting of old ideas as new.

You know, like when you have to sell your car – an analogy I'm using because I just sold mine. You clean it and polish it, give it a good wash, touch up any scratches and make it look as brand new as possible. Some people come along and kick the tyres, pretending they know what they're doing, and are impressed by the gleam, while the experienced ones immediately pop the hood and turn it on and go about their business of giving it a proper once over. No point in a gleaming outside if the inside is a lemon, is there?

That's what this month's cover story is about. While you will get bombarded by shiny exteriors of AI news almost every week in 2018, we're attempting to look under the hood and lay bare the truth of the innards.

Personally, I think most of us tech users are only genuinely interested in AI that mimics human-like aptitude, not attitude. We don't need our AI to be emotional, or to be our friends, or to be anything like us, really; but we expect them to be smart. We expect them to be able to learn, to not repeat

mistakes. We might not even need to see them, and perhaps they could be like Asimov's fictitious MultiVac – an AI superintelligence that's never seen, but is always there nonetheless. An internet connected superintelligence... perhaps even what the Internet itself will become in the future... who knows? Maybe every connected device itself will be like a neuron in a human brain. Useless by itself, but capable of solving pretty much anything when fired in the right configuration and order.

I don't think traditional computing can ever result in this capability. With every new research paper I read in the field of quantum biology, the more convinced I am by the theory that our brain is really a super complex, biological quantum computer. In fact, I can see why some theorists believe that consciousness itself is the result of all the weird effects of quantum mechanics. And let's be honest, for a human-like AI to exist (aka an AGI – read Mithun's story on page 79 to learn more about it), we're going to need artificial consciousness, because without it, we're not going to get anywhere close to human aptitude.

If the quantum biologists are right, and consciousness is in fact a quantum property of the brain, then we are certainly going to need proper quantum computers to be developed and we're going to need software that can be run on such computers. No point in running regular code on a quantum computer. That's like asking an advanced mathematician to do basic addition and subtraction. Of course she'd be able to do it, and do it far faster than a regular adult, but should we be wasting our time with that? What that software will be or look like, we have no idea, because we don't know what the hardware will be... because we still don't fully understand quantum mechanics... a vicious circle of ignorance.

The only thing about this return of the tech world's focus on AI that I like is that new research projects will get some much-needed funding, because of all this hype. New research is where the breakthroughs will come from... so wake me when there's one in quantum computing, or in quantum mechanics; because until then, I think we're doomed to merely interface with dumb Siri-like "AI".

Here's wishing you a very Happy 2018 ahead – just steer clear of "smart assistants" and you the year will be happy enough! 



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